



Gosford High School

Year 11

2009

Preliminary

Higher School Certificate

Mathematics

Assessment Task 2

Time Allowed - 60 minutes

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.
- * untidy and/or poorly organised solutions may not gain full marks.

Question 1 (13 marks)**START A NEW PAGE**

- (a) Find the value $\sqrt{6.714 \times 10^{-3}}$ giving your answer correct to 3 significant figures (1)
- (b) A pump empties water from a tank at a rate of 300L/h.
- (i) How long will it take to empty a tank containing 750L? (1)
- (ii) How much water will it pump in 1 hour 40 minutes? (1)
- (c) Simplify $\frac{2\sqrt{24} - \sqrt{54}}{2\sqrt{3}}$, writing your answer in exact form (2)
- (d) Find a and m if $a + m\sqrt{3} = (3\sqrt{6} - 5\sqrt{2})^2$ (3)
- (e) The average distance from the Earth to the Sun is 1.50×10^8 km.
How long would it take light travelling at 3.0×10^8 m/s to reach the Earth from the Sun.
Give your answer correct to the nearest minute. (2)
- (f) Simplify $\frac{7}{10 - 6\sqrt{2}}$ expressing your answer with a rational denominator. (3)

Question 2 (13 marks)**START A NEW PAGE**

- (a) Expand and simplify $(4x + 3y)(2x - 5y)$ (1)
- (b) Factorise $8x^3 - 1$ (1)
- (c) Factorise fully $36m^2 - 4$ (2)
- (d) Simplify $\frac{1}{2n^2 + n} + \frac{1}{2n^2 + 3n + 1}$ (3)
- (e) Factorise fully then simplify $\frac{a^4 - 1}{a^3 - a^2 - 1 + a}$ (3)
- (f) Simplify $\frac{6^{4n} \times 9^{1-n}}{12^{2n}}$ (3)

Question 3 (18 marks)**START A NEW PAGE**

- (a) Solve $1 \cdot 2a - 0 \cdot 36 = 2 \cdot 4a$ (2)
- (b) Solve $4 - 3y > \frac{y}{2}$ (2)
- (c) Solve $(x-1)^2 = 2$ (2)
- (d) If -3 is a root of the equation $x^2 + kx - 6 = 0$, find the value of k . (2)
- (e) Graph on a number line the set of real values of x satisfying $x > 2$ OR $x \leq -1$ (2)
- (f) Solve $2x^2 - 6x - 1 = 0$ writing your answers correct to 2 decimal places (3)
- (g) If $S = \frac{a(1-r^n)}{1-r}$ find the exact value of S when $a = 32$, $n = 8$ and $r = -\frac{1}{2}$ (2)
- (h) Solve simultaneously $\begin{cases} 3x + 7y = 27 \\ 5x + 2y = 16 \end{cases}$ (3)

Question 4 (16 marks)**START A NEW PAGE**

- (a) Solve $|2x-1| \geq 3$ (2)
- (b) Solve $9^x = \frac{1}{27}$ (2)
- (c) Solve $2x^2 - 5x \leq 0$ (3)
- (d) Solve $|2x-4| = 7x+1$ (3)
- (e) Solve simultaneously $\begin{cases} y - 2x = 1 \\ x^2 + y^2 = 10 \end{cases}$ (4)
- (f) Make y the subject of the formula $y^2 - 4xy + 1 = 0$ (2)

Question 1 Prelim Task 2

a) 0.0819

b) i) 2.5 hours

ii) 500 L

$$\begin{aligned} c) \frac{2\sqrt{24} - \sqrt{54}}{2\sqrt{3}} &= \frac{4\sqrt{6} - 3\sqrt{6}}{2\sqrt{3}} \\ &= \frac{\sqrt{6}}{2\sqrt{3}} \\ &= \frac{\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} d) R.H.S. &= 54 + 50 - 30\sqrt{12} \\ &= 104 - 60\sqrt{3} \\ \therefore a &= 104, m = -60 \end{aligned}$$

$$\begin{aligned} e) \text{Time} &= \frac{\text{Distance}}{\text{Speed}} \\ &= \frac{1.50 \times 10^{11}}{3 \times 10^8} \text{ seconds} \\ &= 500 \text{ seconds} \\ &\approx 8 \text{ minutes} \\ &\text{(to nearest minute)} \end{aligned}$$

$$\begin{aligned} f) \frac{10 - 6\sqrt{2}}{10 - 6\sqrt{2}} &= \frac{10 - 6\sqrt{2}}{10 + 6\sqrt{2}} \\ &= \frac{7(10 + 6\sqrt{2})}{100 - 72} \\ &= \frac{14(5 + 3\sqrt{2})}{28} \\ &= \frac{5 + 3\sqrt{2}}{2} \end{aligned}$$

Question 2

a) $8x^2 - 14xy - 15y^2$

b) $(2x - 1)(4x^2 + 2x + 1)$

c) $4(9m^2 - 1) = 4(3m - 1)(3m + 1)$

$$\begin{aligned} d) \frac{1}{n(2n+1)} + \frac{1}{(2n+1)(n+1)} \\ &= \frac{n+1 + n}{n(2n+1)(n+1)} \\ &= \frac{2n+1}{n(2n+1)(n+1)} \\ &= \frac{1}{n(n+1)} \end{aligned}$$

$$a^2(a-1) + 1(a-1)$$

$$= \frac{(a-1)(a+1)(a^2+1)}{(a^2+1)(a-1)}$$

$$= a+1$$

$$\begin{aligned} f) \frac{(3 \times 2)^{4n} \times (3^2)^{1-n}}{(2^2 \times 3)^{2n}} \\ &= \frac{3^{4n} \times 2^{4n} \times 3^{2-2n}}{2^{4n} \times 3^{2n}} \\ &= 3^{2n} \times 3^{2-2n} \\ &= 3^2 \\ &= 9 \end{aligned}$$

Question 3

a) $-0.36 = 1.2a$

$a = -0.3$

b) $8 - 6y \geq y$

$8 > 7y$

$y < \frac{8}{7}$

c) $x - 1 = \pm\sqrt{2}$

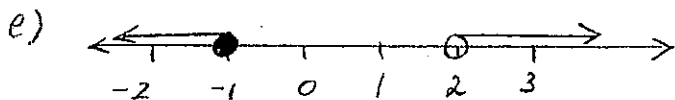
$x = 1 \pm \sqrt{2}$

d) $(-3)^2 + k(-3) - 6 = 0$

$9 - 3k - 6 = 0$

$3 = 3k$

$k = 1$



f) $x = \frac{6 \pm \sqrt{36 - 4(2)(-1)}}{4}$

$x = \frac{6 \pm \sqrt{44}}{4}$

$x \approx 3.16, -0.16$

g) $S = \frac{32(1 - (-\frac{1}{2})^8)}{1 + \frac{1}{2}}$

$S = \frac{2}{3} \times 32 \times (1 - \frac{1}{256})$

$= \frac{2}{3} \times 32 \times \frac{255}{256}$

$= 21\frac{1}{4}$

h) $6x + 14y = 54$

$35x + 14y = 112$

$29x = 58$

$x = 2$

$\therefore 6 + 7y = 27$

$7y = 21$

$y = 3$

Question 4

a) $2x - 1 \geq 3$ or $2x - 1 \leq -3$

$2x \geq 4$

$2x \leq -2$

$x \geq 2$

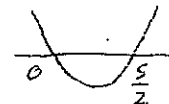
$x \leq -1$

b) $(3^2)^x = 3^{-3}$

$2x = -3$

$\therefore x = -\frac{3}{2}$

c) $x(2x - 5) \leq 0$



$\therefore 0 \leq x \leq \frac{5}{2}$

d) $2x - 4 = 7x + 1$ ~~or~~ $2x - 4 = -7x - 1$

$-5 = 5x$

$9x = 3$

$-1 = x$

$x = \frac{1}{3}$

Does not satisfy

Satisfies

e) $y = 1 + 2x$

$\therefore x^2 + (1 + 2x)^2 = 10$

$x^2 + 1 + 4x + 4x^2 = 10$

$5x^2 + 4x - 9 = 0$

$(5x + 9)(x - 1) = 0$

$\therefore x = -\frac{9}{5}, 1$

$y = -\frac{13}{5}, 3$

f) $y = \frac{4x \pm \sqrt{(4x)^2 - 4(1)(1)}}{2}$

$y = \frac{4x \pm \sqrt{16x^2 - 4}}{2}$

$= 2x \pm \sqrt{4x^2 - 1}$